

A/B Test Report

Landing-page conversion experiment

Requested pricing-study deliverable — analysed as conversion data

CONTROL CONVERSION

12.039%

Old landing page

TREATMENT CONVERSION

11.881%

New landing page

ESTIMATED CHANGE

-0.158 pp

Treatment minus control

Executive decision

Do not roll out the new landing page on the basis of conversion. The treatment estimate is -0.158 pp relative to control and the two-sided p-value is 0.190. The result does not provide statistically significant evidence of a conversion improvement at $\alpha = 0.05$.

Scope note

The CSV contains user assignment, landing page, timestamp and conversion only. It has no price, revenue, margin or customer-segment fields. Accordingly, this report tests landing-page conversion only and makes no claim about pricing impact or profitability.

Prepared for

Advik Singh

Reg. No. RA2411056030023 | Branch: DS-A

Report date: 28 September 2026

Observation window: 02 Jan 2017 – 24 Jan 2017

Analysis population: 290,584 unique users after quality controls

Experiment design & data quality

Pre-analysis checks ensure that each retained user belongs to one intended experimental condition.

Design

Unit of randomisation: user ID. Control users were assigned the old page; treatment users were assigned the new page. The primary metric is the binary conversion indicator. The primary comparison is treatment conversion minus control conversion using a two-sided test at $\alpha = 0.05$.

Cleaning audit

Check	Count	Treatment
Raw records read	294,478	Rows in the supplied CSV
Missing values	0	Across required fields
Malformed records	0	Excluded before assignment checks
Assignment/page mismatches	3,893	Control/new or treatment/old
Rows after alignment	290,585	Valid intended exposures
Repeated-user records removed	1	2 rows across 1 user(s); earliest retained
Final analysis population	290,584	One aligned record per user

Allocation check

Arm	Users	Share
Control (old page)	145,274	49.99%
Treatment (new page)	145,310	50.01%

SAMPLE-RATIO CHECK

p = 0.947

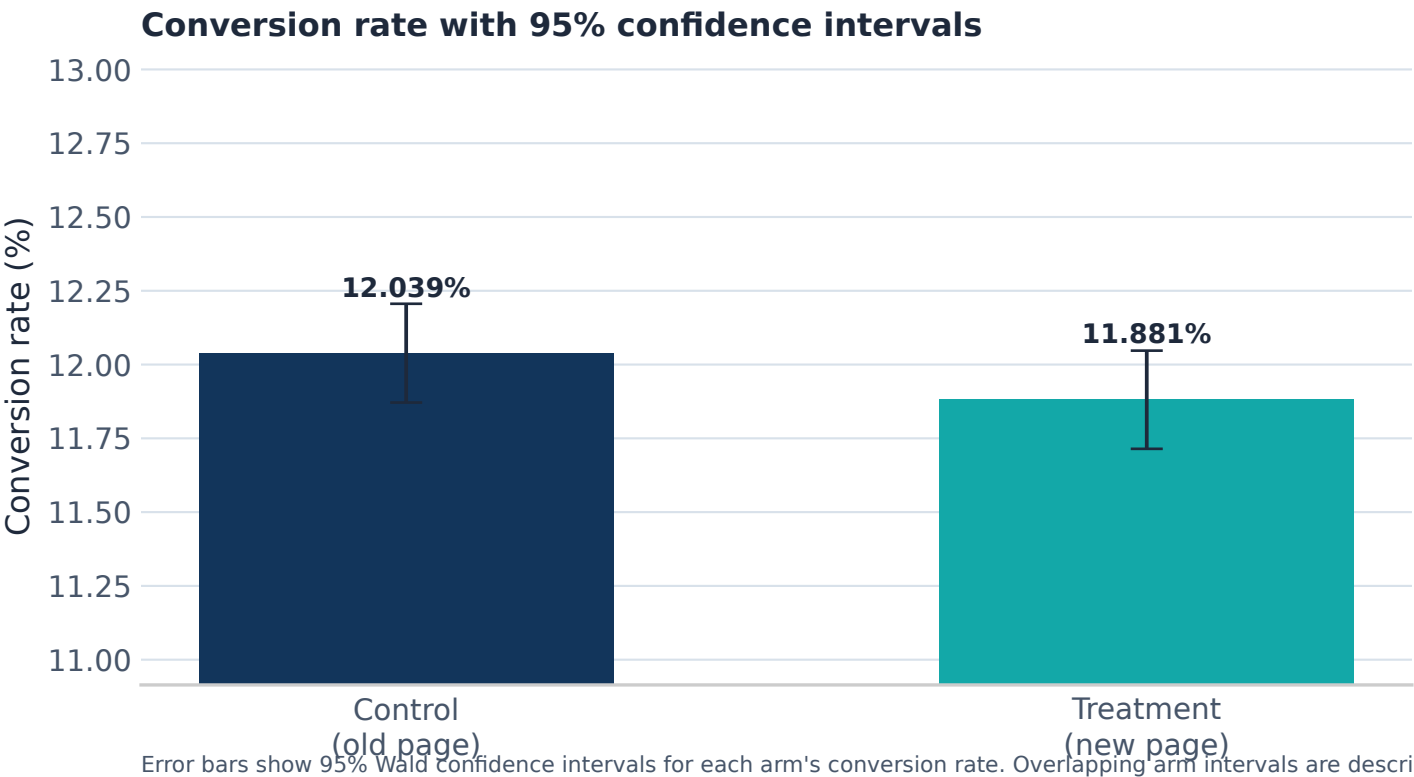
Chi-square = 0.004; expected 50/50

Quality conclusion

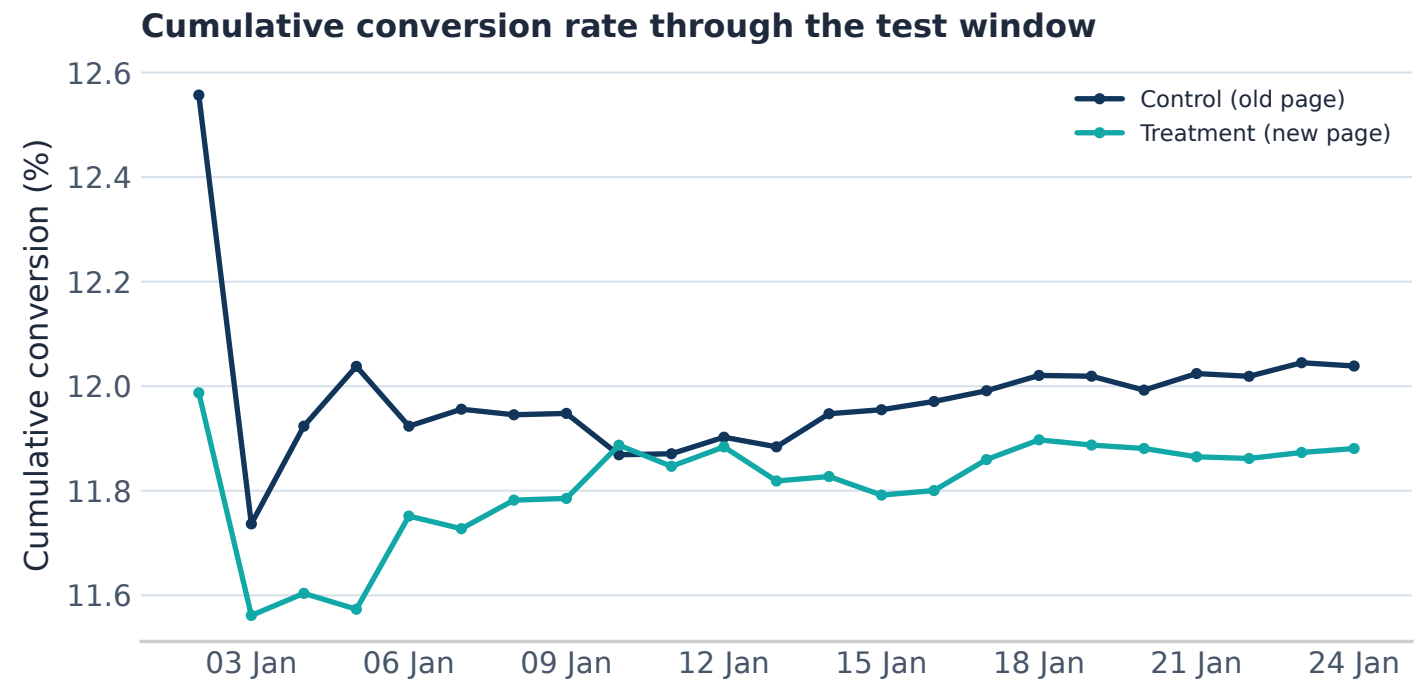
The retained sample is essentially evenly allocated, with no evidence of a sample-ratio mismatch. Removing assignment/page contradictions and repeated users is necessary to compare the intended new-page and old-page experiences. No customer covariates are available, so baseline balance beyond arm counts cannot be assessed.

Visual evidence

The treatment line remains below the control line over the observed period; visual separation is small.



Error bars show 95% Wald confidence intervals for each arm's conversion rate. Overlapping arm intervals are descriptive; the formal difference test appears on the next page.



Cumulative rates are descriptive monitoring only. The experiment's primary decision uses the full-window comparison, not repeated daily significance testing.

Primary inference: new page vs. old page

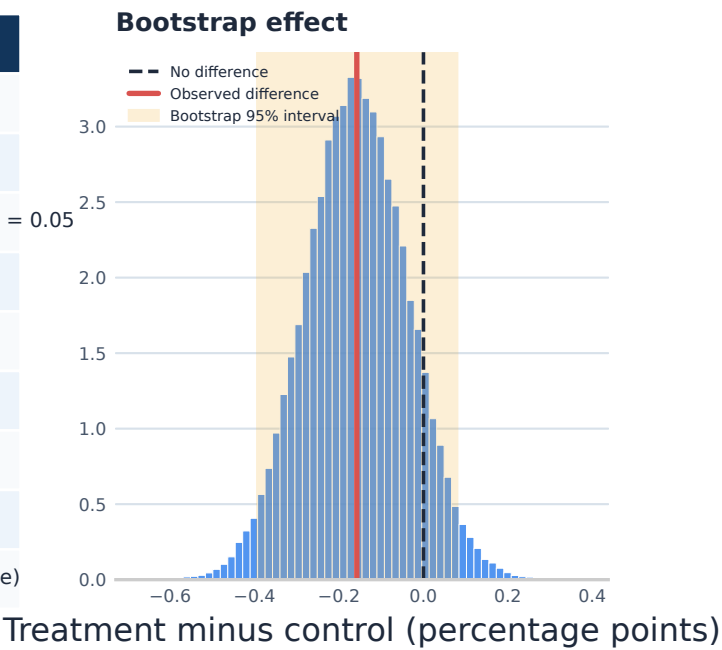
Two independent proportions; treatment effect is always defined as new-page minus old-page conversion.

Observed outcomes

Arm	Users	Conversions	Conversion rate
Control — old page	145,274	17,489	12.039%
Treatment — new page	145,310	17,264	11.881%

Hypotheses and test

Measure	Result
Null hypothesis	$H_0: p(\text{new page}) - p(\text{old page}) = 0$
Alternative hypothesis	$H_1: p(\text{new page}) - p(\text{old page}) \neq 0$
Test and threshold	Two-proportion z-test, two-sided $\alpha = 0.05$
Test statistic	$z = -1.311$
Two-sided p-value	0.1899
95% CI for difference	$[-0.394 \text{ pp}, +0.078 \text{ pp}]$
Bootstrap 95% interval	$[-0.393 \text{ pp}, +0.080 \text{ pp}]$
Relative conversion change	-1.31%
Standardised effect (Cohen's h)	-0.0049 (negligible practical magnitude)



Interpretation

Because $p = 0.1899$ is greater than 0.05, the analysis does not reject the null hypothesis. The 95% interval ranges from a 0.394 percentage-point treatment loss to a 0.078 percentage-point gain. It is therefore not evidence that the new page improves conversion; it is also not proof that the two pages are exactly equivalent.

Power, practical significance & recommendation

Statistical non-significance should be interpreted alongside the size of the effect the experiment could reliably detect.

Sensitivity of the experiment

OBSERVED ABSOLUTE CHANGE

0.158 pp

Magnitude of the estimated difference

80% POWER MDE

0.340 pp

Two-sided alpha = 0.05

NEEDED PER ARM

671,133

To detect observed magnitude at 80% power

With about 145,274 users per arm, this study has 80% power to detect an absolute conversion change of roughly 0.340 pp or larger. The observed absolute difference is 0.158 pp, about 46% of that threshold. A future experiment targeting an effect this small would need approximately 671,133 users in each arm under the same baseline-rate assumptions.

Recommendation

Do not adopt the new page as a conversion improvement.

The treatment point estimate is lower and the confidence interval includes zero. Retain the current page for the conversion objective, or continue a pre-registered experiment only if the business considers a small conversion difference decision-relevant. A conversion-neutral rollout cannot be established from a non-significant result alone.

Actions before a follow-up test

- Pre-specify a minimum practical effect (MDE) jointly with product stakeholders, then size the experiment for that target rather than for the observed result.
- Capture revenue, price, margin, refunds and guardrail metrics (for example, latency or support contacts) if the decision is genuinely about pricing or profitability.
- Keep one primary conversion metric, a fixed stopping rule and a documented duration; treat any subgroup or daily views as exploratory unless they are pre-registered and multiplicity-controlled.

Methodology, provenance & reproducibility

All calculations are reproducible from the source CSV and the included Python script.

Methods

- Cleaning: require the five expected fields; retain only control/old-page and treatment/new-page assignments; retain the earliest valid row for any repeated user ID.
- Primary estimate: treatment conversion rate minus control conversion rate. Significance test: pooled two-proportion z-test, two-sided alpha = 0.05.
- Uncertainty: 95% unpooled Wald confidence interval plus a 100,000-replicate binary non-parametric bootstrap (seed 42).
- Power: normal-approximation, equal-sized arms, 80% power and two-sided alpha = 0.05. It is planning context, not a post-hoc claim of achieved power.

Data provenance

Item	Value
Dataset	ab_data.csv
Original repository	aamir-gk2539-byte/DSML-Training
Source file	github.com/aamir-gk2539-byte/DSML-Training/blob/main/ab_data.csv
Source commit	94acc3eb1ec8
SHA-256	d56e2acce25e99ac21cb3d76c5df516dd19cc7a77c14c9014f94e1ea1301beb
Retrieved / report date	28 September 2026

Reproduce

```
pip install -r requirements.txt    # then:    python AB_Test_Analysis.py
```

Limitations

This observational extract identifies assignment and conversion but does not document randomisation implementation, exposure quality, business costs, price, revenue, user characteristics or longer-term outcomes. Results generalise only to the observed test window and the recorded conversion event.